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Disentangling climate signals from survey effort in new bird distribution records across China

Birds or birders? A discrete-time hazard model of first records, 2002–2024

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Background

A new record is the last link in a chain of three events

New distribution record: the first documented occurrence of a species in a province where it had not been recorded before

1 The bird is there



The species occupies the province; warming can move ranges

Ecological process

2 Someone looks



A birder or a survey detects it; effort grew fast after 2000

Observation process

3 The find is published



The detection is written up and dated in a paper or report

Documentation

New record

Only this link is ecological

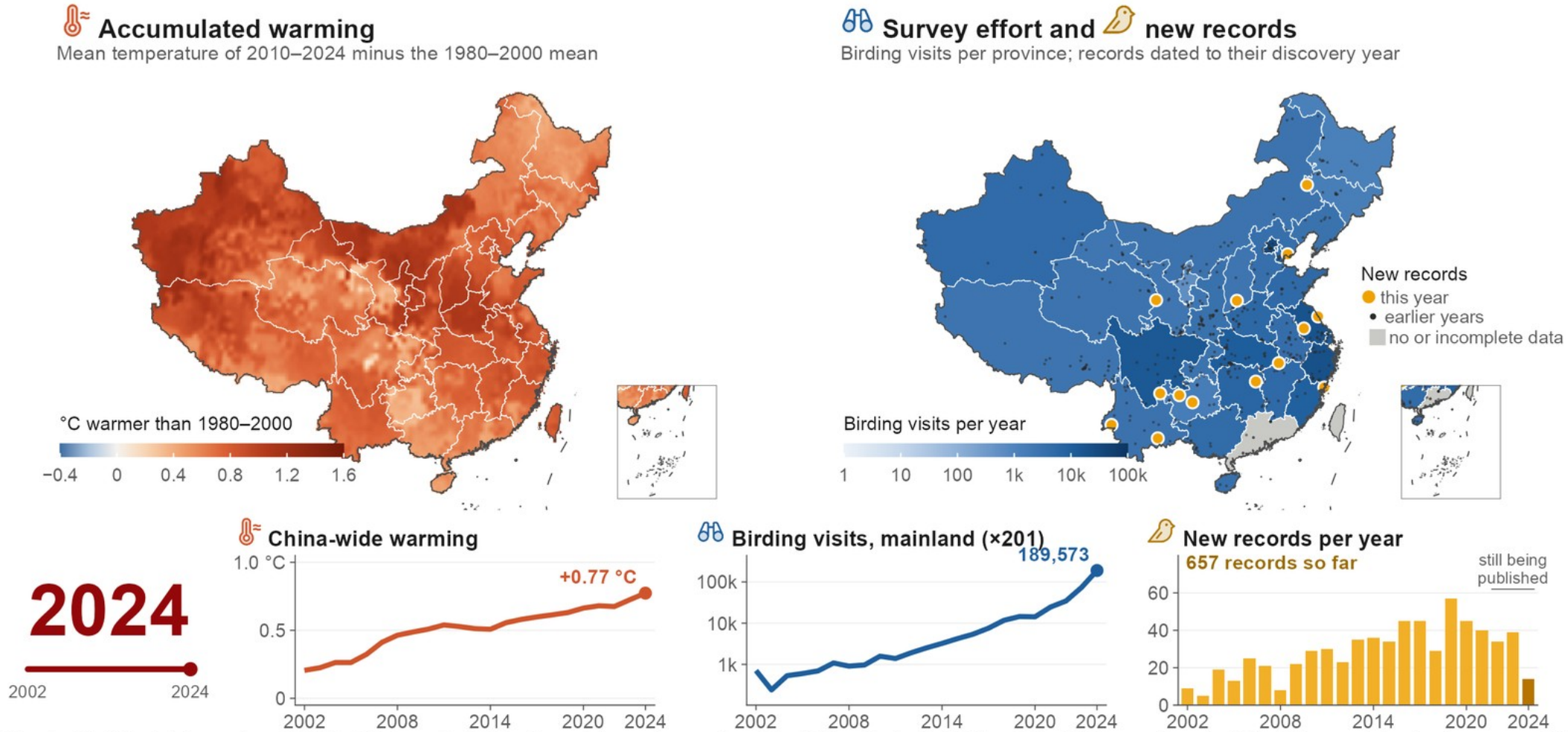
These two leave the same trace in the data

- ☐ Imperfect detection and uneven sampling can create apparent range changes where none occurred
- ☐ In China, provincial new bird records have been read as evidence of poleward shifts

Tingley & Beissinger 2009; Hughes et al. 2021; Xing et al. 2024; Chen et al. 2025

Background

After 2002, warming, birding and new records all rose together



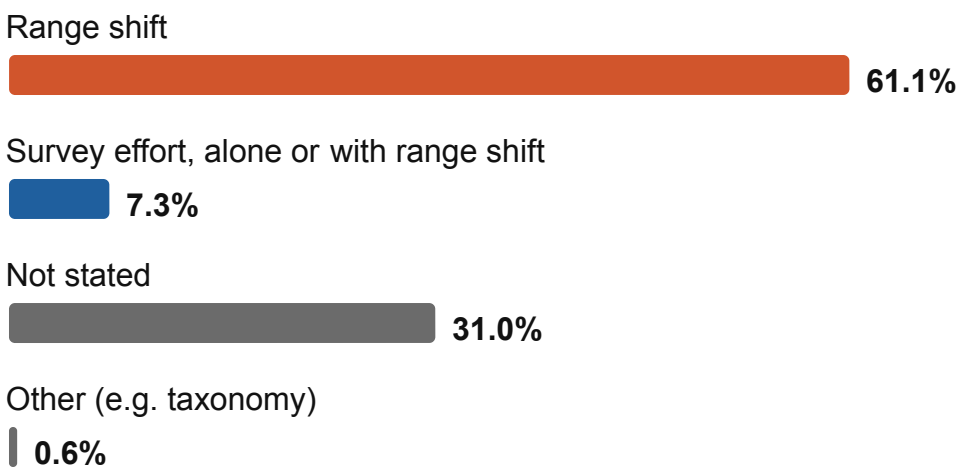
Warming: WorldClim 2.1 downscaling of CRU TS 4.09, the provincial side of the model's warming covariate. Effort: birding visits in eBird/GBIF and China Bird Report. Records: CBNR v2 first records dated to discovery, n = 657. Grey: effort coverage gap; outside the analysis (Taiwan, Hong Kong, Macao); or incomplete download (Guangdong 2024). Base map: GS(2019)1822.

Research gaps

Analyses of new records have not separated ecology from observation

How source papers explain their records

CBNR v2, n = 1,029 provincial first records



Birding visits rose about 200-fold, 2002–04 to 2022–24, yet effort is rarely given as the reason

1 GAP — Confounded drivers

Warming and observation grew together; a model with only one of them credits it with both

2 GAP — Unmodelled observation

Survey pulses, and which species could be recorded where, are rarely in the model

3 GAP — No benchmark, no support check

Warming lacks a null benchmark; projections are rarely checked against the fitted data

Needed: one model that treats each record as the product of ecology and observation

Questions

Four questions, each with a testable expectation

Q1 Does warming raise the rate of new records once observation is modelled?

H1 Yes, beyond the benchmark that shared regional climate alone produces

Q2 How large is the observation process itself?

H2 Province-year survey pulses are the largest source of variation

Q3 Do warming and survey effort interact?

H3 Negatively: where effort is high, species favoured by warming are already recorded

Q4 Are the returns on warming and on effort stable through time?

H4 Returns on effort decline as gaps are filled; returns on warming do not

Approach

From records to a hazard model, in four steps

1 Records

- **CBNR v2** — 1,029 provincial first records
- **Dated to discovery** — 657 records in 2002–2024
- **Current release, v1.1** — 1,076 records · 588 species

2 Risk set

- **Species × province × year** — 175,640 cells, 2002–2024
- **Reachable provinces only** — ≥ 50 suitable SDM cells
- **Exit after first record** — each pair counted once

3 Covariates

- **Warming** — vs species range, 15 yr
- **Survey effort** — birding visits
- **Variability** — annual anomaly
- **Completeness** — offset, recent years

4 Analysis

- **Hazard GLMM** — cloglog link
- **Robustness** — nulls · 12 specs
- **Through time** — time-varying returns
- **Projections** — support checked

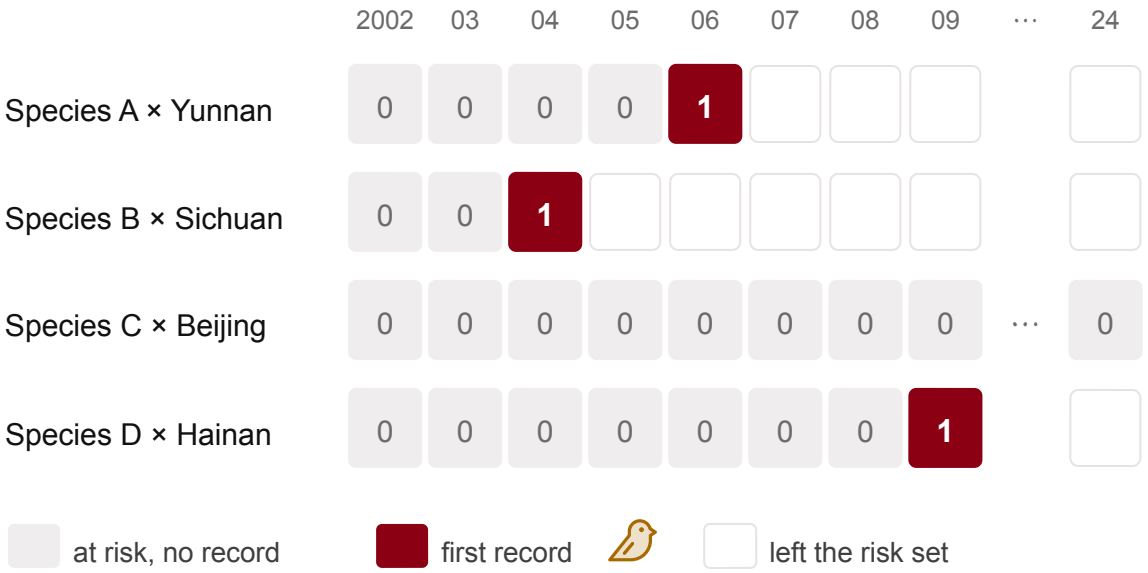
Outputs: contributions · interaction · observation process · returns through time · futures

Approach

A discrete-time hazard model asks: is this the year of the first record?

The risk set

Each pair is at risk every year until its first record



175,640 cells · 649 first records

392 species · 31 provinces · 2002–2024

first record ~ warming × effort + variability
+ offset(log completeness)

+ (1 | species) + (1 | province) + (1 | province × year)

Binomial, cloglog link: exp(β) is a hazard ratio per SD

Accumulated warming

Province minus species-range anomaly, 15-yr; 1 SD = 0.18 °C

Survey effort

Provincial birding visits; 1 SD = 7.9-fold more visits

Province × year effect

Survey campaigns, birding festivals, new reserves

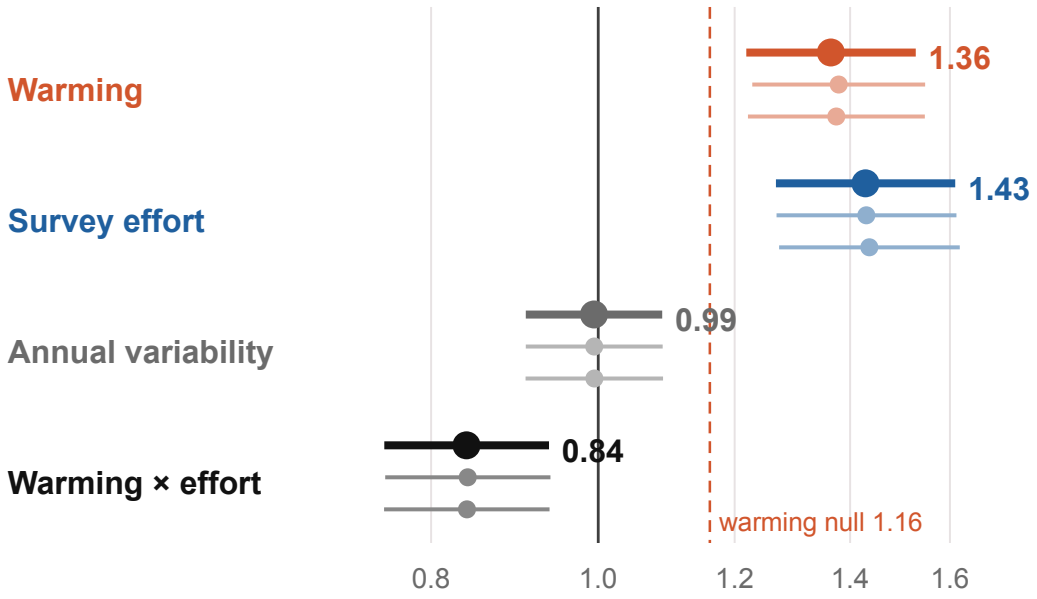
Dated to discovery

Offset corrects the latest, incompletely published years

Finding 1

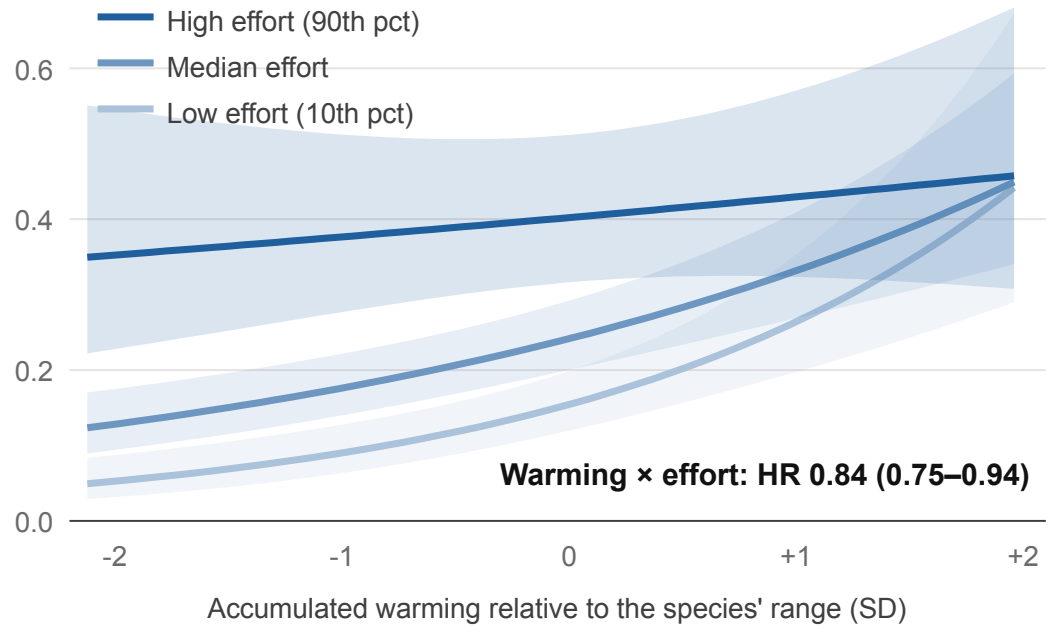
Warming and survey effort contribute comparably, and they interact

Hazard ratio per SD (95% CI)



Thick: main analysis (≥ 50 suitable SDM cells); thin: 100 and 200 cells; log scale

Annual hazard of a first record (%)



Accumulated warming
1.36
HR per SD (1.22–1.53)

Survey effort
1.43
HR per SD (1.27–1.61)

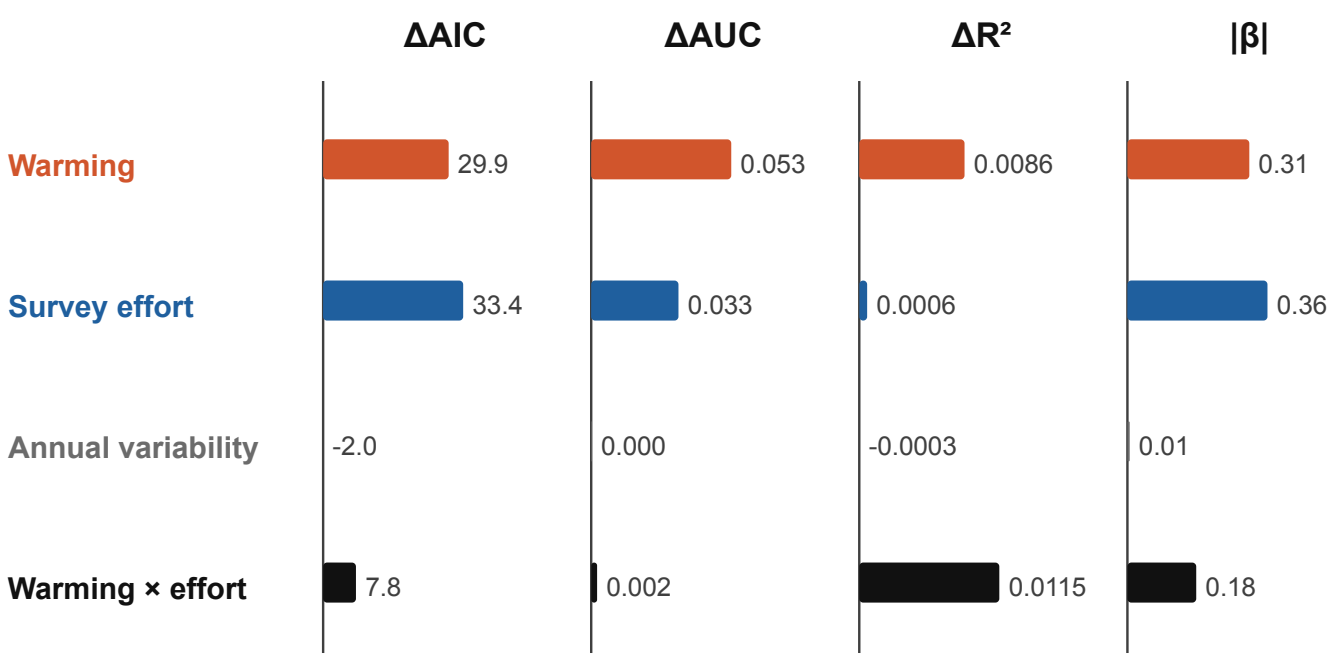
Warming × effort
0.84
steepest where few look

Annual variability
0.99
no effect (0.91–1.09)

Finding 1

The balance holds on average across criteria and twelve specifications

Loss in fit when the term is dropped



Each column scaled to its own maximum; |β|: absolute standardised coefficient

Warming and effort rank equally on average across the four criteria

Cost of dropping a term

ΔAIC: effort 33.4, warming 29.9
ΔAUC: warming 0.053, effort 0.033
|β|: effort 0.36, warming 0.31

Twelve specifications

SDM thresholds 50, 100, 200
4 effort proxies × 3 missing-data rules
Effort 1.25–1.55; warming 1.33–1.39

Design-preserving nulls

Warming null 1.16: increment ×1.18
Shuffled effort timing: 0.98
Province bootstrap: lower bounds > 1

Finding 2

Survey pulses are the largest source of variation in the hazard

Observation shocks vary by province and year

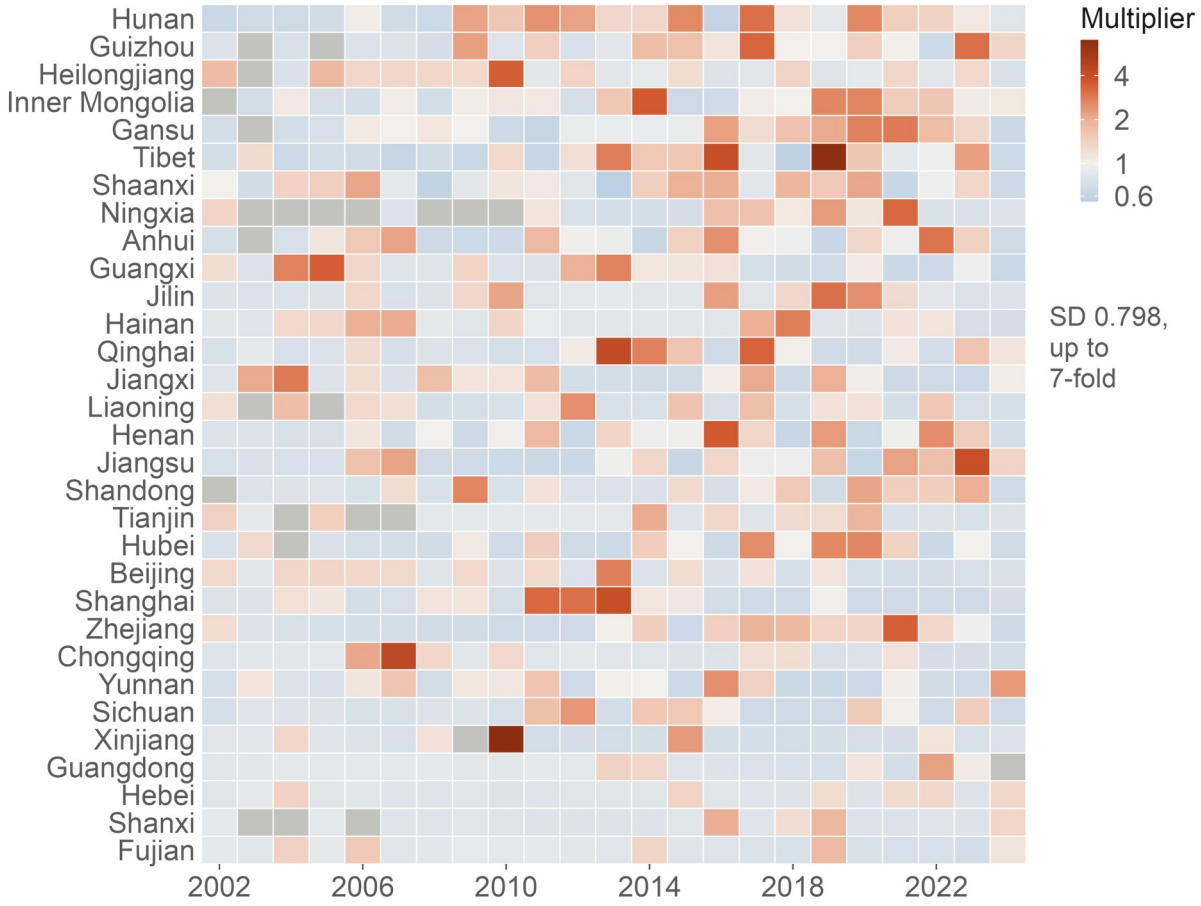
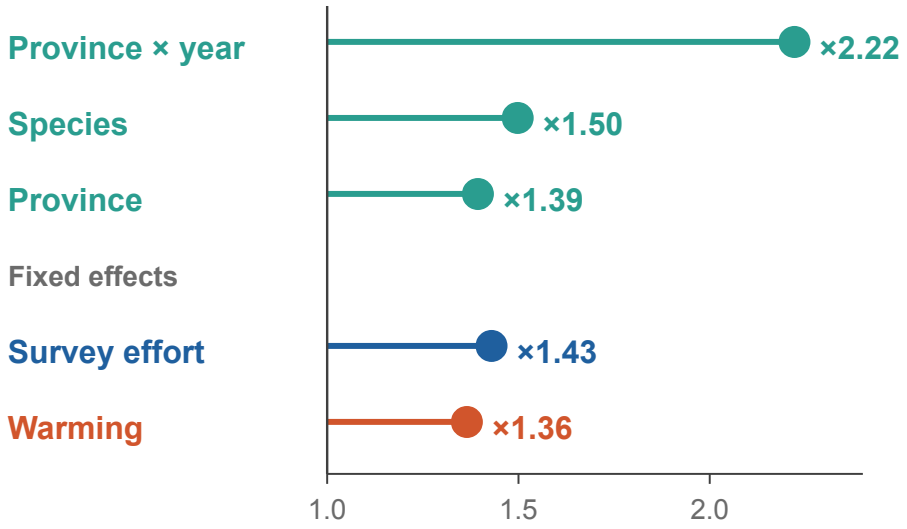


Fig. 5c Province × year multipliers

Hazard multiplier per SD



Conditional AUC

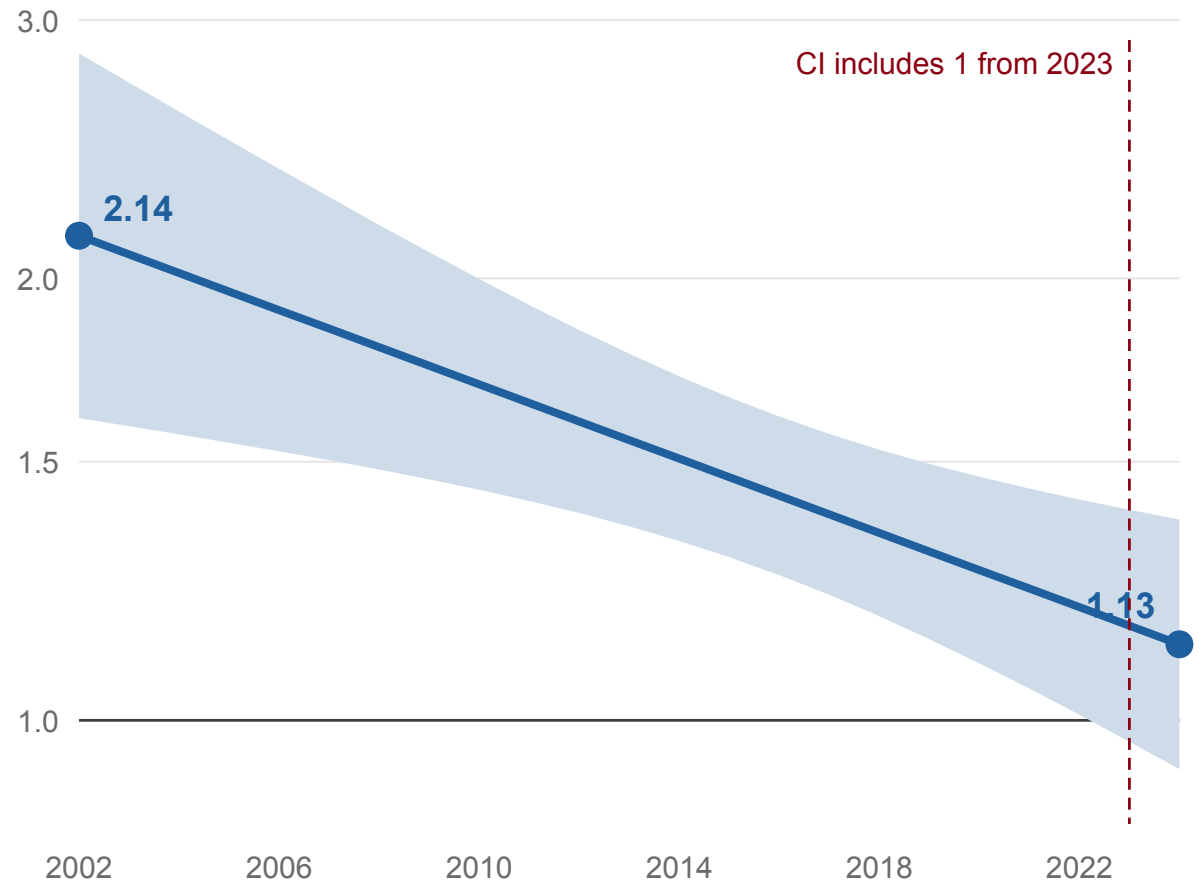
0.72 → 0.85

after adding the province × year layer

Finding 3

The return on effort is falling; the return on warming is not

Return on survey effort: HR per SD (95% CI)



Return on effort

2.14 → 1.13

HR per SD, 2002 → 2024; P = 0.0021

Return on warming

P = 1.00

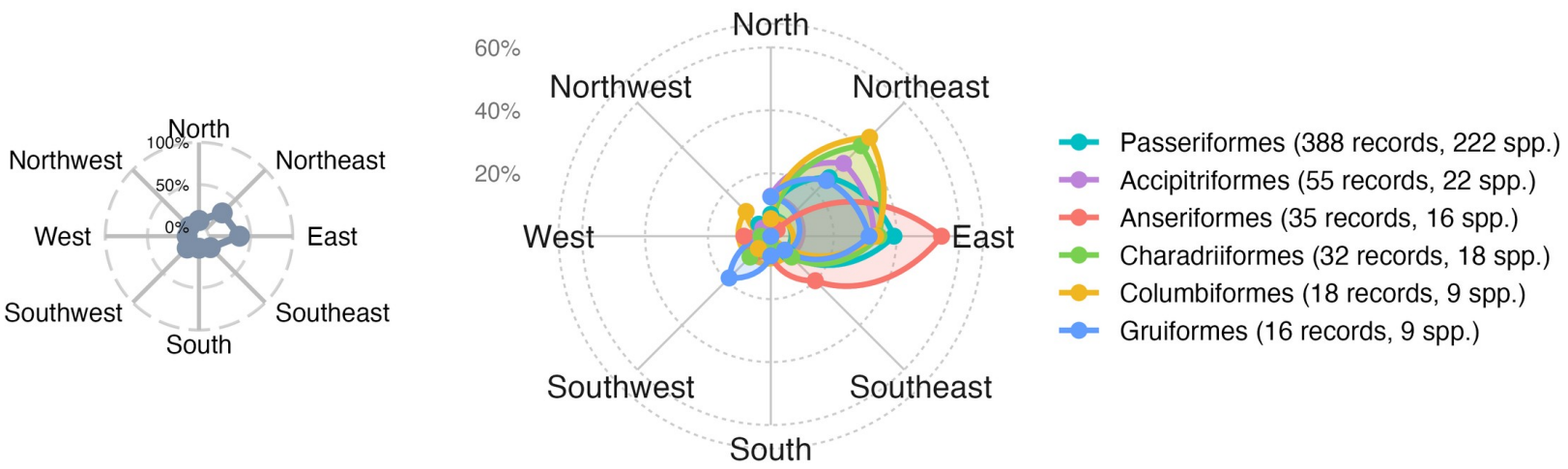
warming × time: no trend; HR 1.37

A stock and a flow

Effort mines a finite stock of undocumented occurrences; warming keeps adding new candidates

Finding 4

Seen from range centres, new records point east and north-east



Share of each order's own new records; absolute sample sizes are given in the legend

Bearings from range centroids: all records (left), six largest orders (right)

Most new records lie east or north-east of the species' range centre

Records

642

361 species, 20 orders

East

36.9%

237 of 642 records

North-east

27.9%

179 of 642 records

Finding 4

The eastward bias is geometry; net of it, records lean slightly north

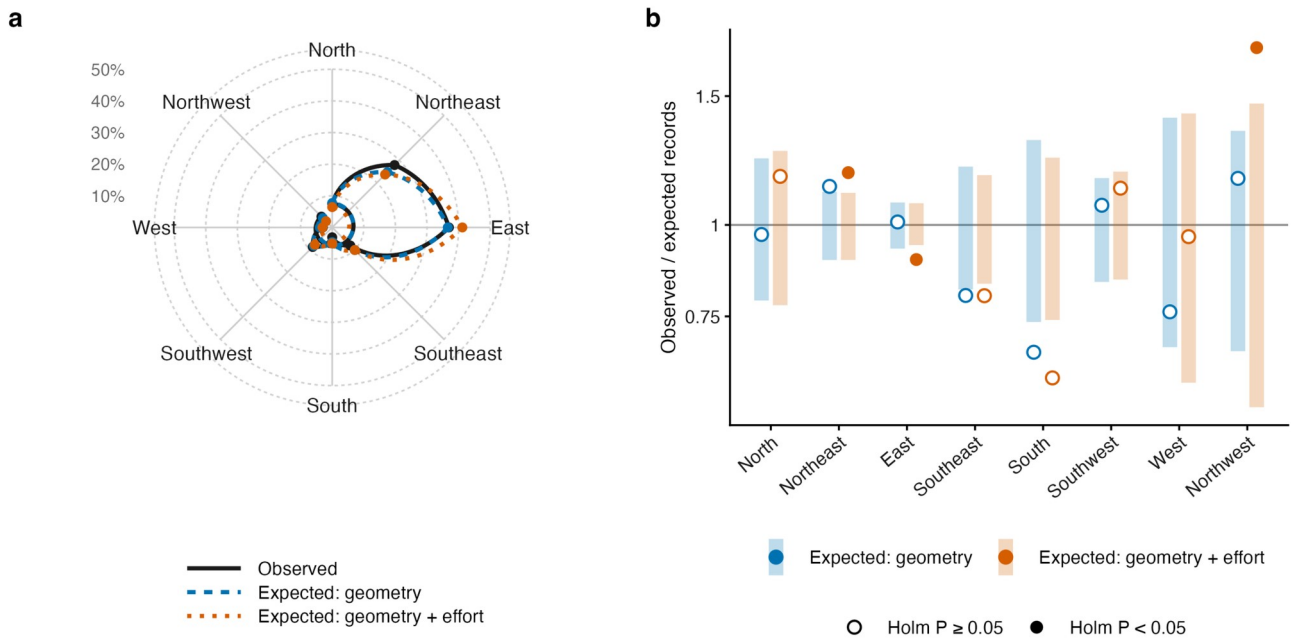


Fig. 6 Observed vs expected sectors under two nulls

- Null 1: each record placed at random among its species' reachable provinces
- Null 2: the same provinces, weighted by birding effort

Null 1: geometry only

East: 235 expected, 237 observed

No sector departs from the null

Null 2: geometry + birders

Northern sectors: 40.2% vs 32.8%

$\chi^2 = 26.8$, $P = 0.001$

NE, NW above; E below (Holm)

Net of geometry and effort: a modest poleward lean

Application

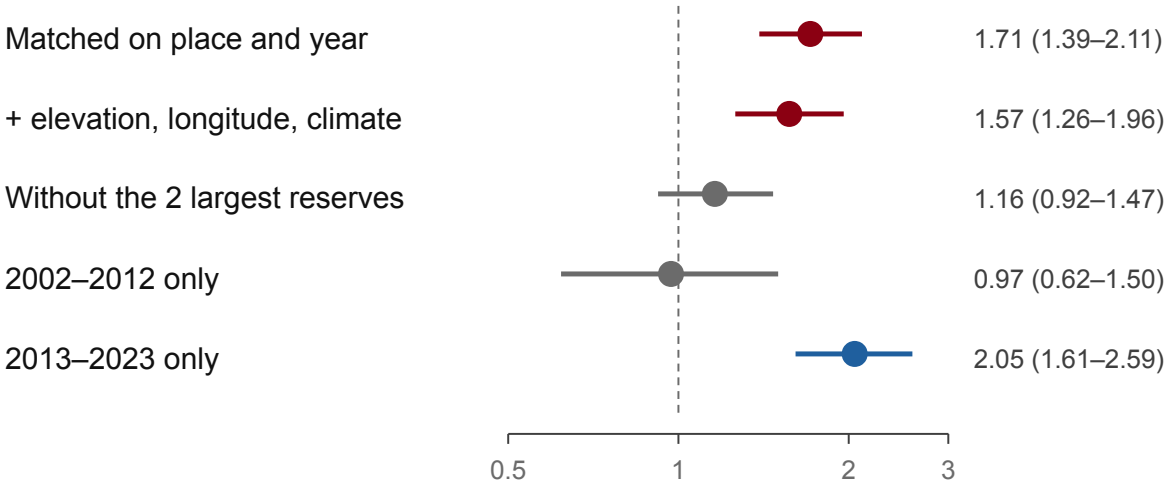
Records inside reserves track observation, not reserve performance

Share inside the 1,028 mapped reserves



Odds ratio against 200 matched controls

Controls from the same province and year, weighted by checklists; 95% CI



Few reserves carry the signal

172 events in 68 reserves

The top two hold 27.3% of them

Without them: OR 1.16

Only after birding boomed

2002–2012: OR 0.97

2013–2023: OR 2.05

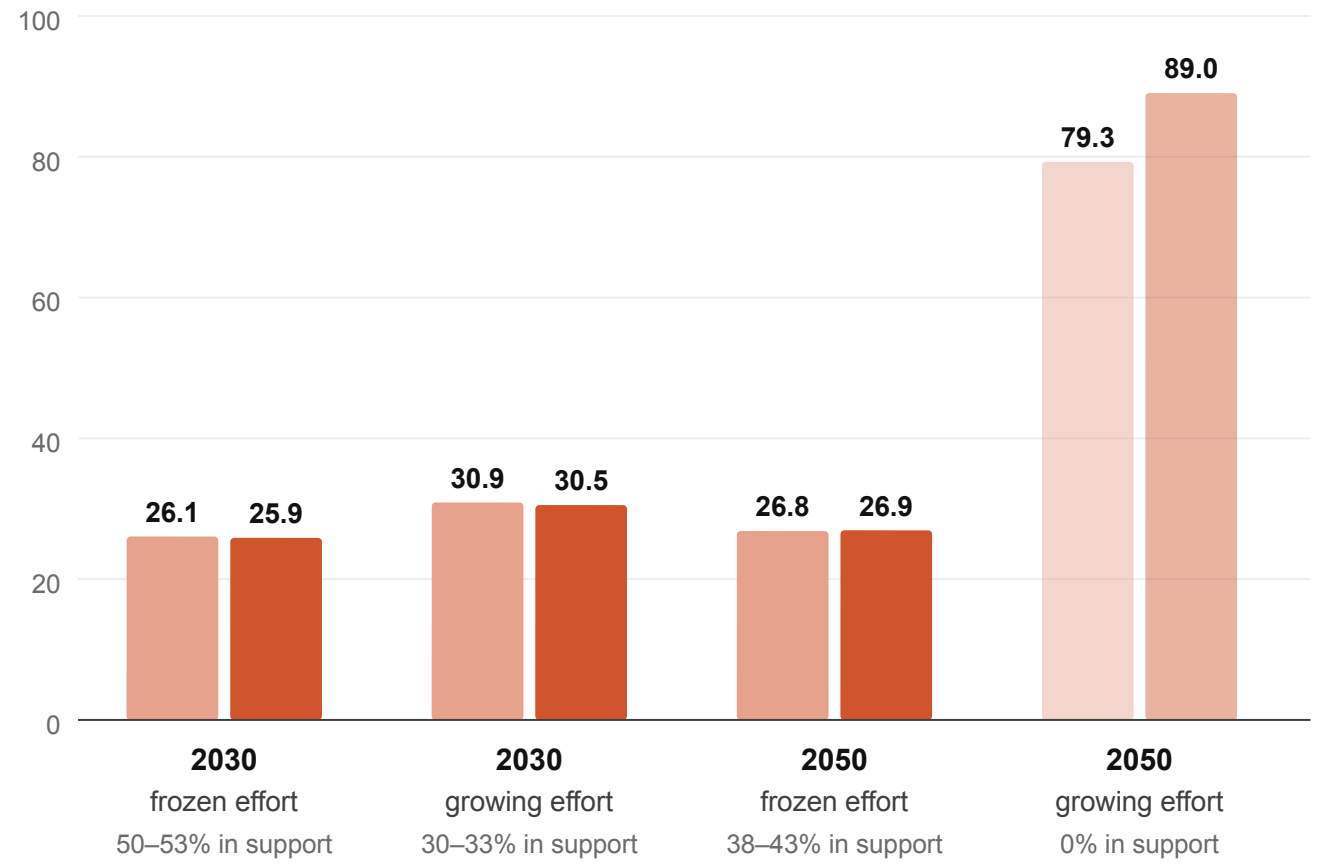
24% more species per visit inside

Record counts should not index reserve performance

Futures

Future record counts follow survey effort, not the emission scenario

Expected new records per year, China



Two climate × two effort scenarios

SSP2-4.5 or SSP5-8.5

Effort frozen, or still growing

What changes the count?

Climate scenario: barely anything

Growing effort: roughly triple

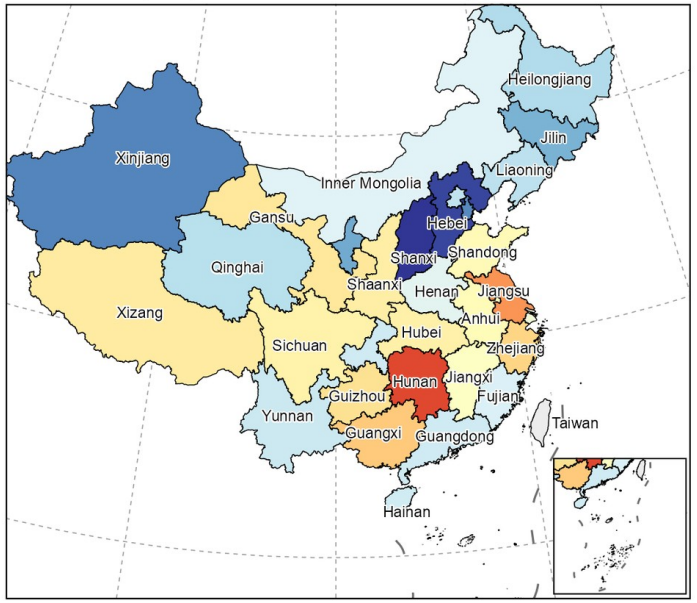
Most of that rise lies outside the fitted data (faded bars: none in support)

Futures

Records spread across central and southern China; growing effort leaves the data

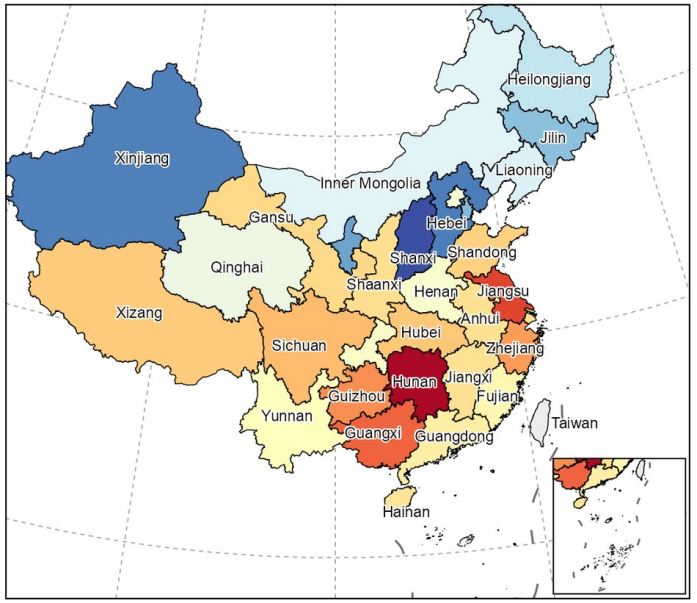
2030 · SSP2-4.5 · effort frozen at 2024

China 26.1 records per year · 53% of rows in support

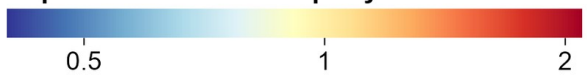


2030 · SSP2-4.5 · effort growing (+0.6 SD per decade)

China 30.9 records per year · 33% of rows in support



Expected new records per year



Percentages: share of species rows inside the fitted support. Grey: no effort data.
Under SSP5-8.5 the two effort scenarios give 25.9 and 30.5 records per year: almost the same.

2030 · SSP2-4.5 · expected new records per province; CBNR map standard, GS(2019)1822

Growing effort adds records in the south and east, beyond the data

Hunan, the top

1.7 → 2.1

per year: frozen → growing

Provinces off-data

8 → 16

< 50% of rows in support

Top five provinces

25%

of the national total

Futures

Data support, not ambition, sets the forecast horizon



Fig. 7b Hazard model vs machine learning

Support shrinks fast

Cells inside the fitted range:

2030: 46% → 2050: 6%

2080: none (SSP2-4.5)

Machine learning does not help

Agree inside the range (ρ = 0.56)

Diverge outside it

2030: usable
End of century:
illustration, not forecast

Limits

Four limits travel with these conclusions

Events are published records, not detections

We explain how records are documented, not range expansion itself.

Coordinates are precise to about 1 km at best

Reserve-overlap tests therefore use 0–5 km buffers as a sensitivity check.

Reserve polygons cover about a third of listed reserves

1,028 mapped reserves: no gap analysis, no 30×30 claims.

Projections leave the fitted support after 2030

Late-century maps are illustrations; the effort coefficient used is an upper bound.

Summary

Warming and survey effort leave signals of similar size

New records carry a climate signal mixed with an observation signal

01 Climate signal

Warming 1.36 vs effort 1.43 per SD: comparable, separable

02 Where few look

Warming × effort 0.84: under-surveyed regions hide the signal

03 Observation process

Province × year ×2.22; returns on effort 2.14 → 1.13

04 Futures

Emission scenario barely matters; most change is extrapolation

Implications

- ☐ Send survey effort to climatically exposed, under-surveyed regions
- ☐ Read counts of new records as observation-weighted evidence, not as range shifts
- ☐ Report where a model leaves its data support before mapping the future

Birds or birders? Both. Model how we look, and new records tell us how ranges change

Thank you

Questions welcome

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Open data and code

CBNR v2 (CC BY 4.0) · doi.org/10.5281/zenodo.20809949

github.com/dingchenchen6/bird-newrecord-discovery-dated

Climate: WorldClim 2.1 / CRU TS 4.09 · Ranges: BirdLife International
Effort: eBird/GBIF and China Bird Report · Base map: GS(2019)1822

We thank birdwatchers across China, and every author whose records made CBNR possible.

CBNR data

Backup

Dating to publication inflates effort and creates a phantom climate effect

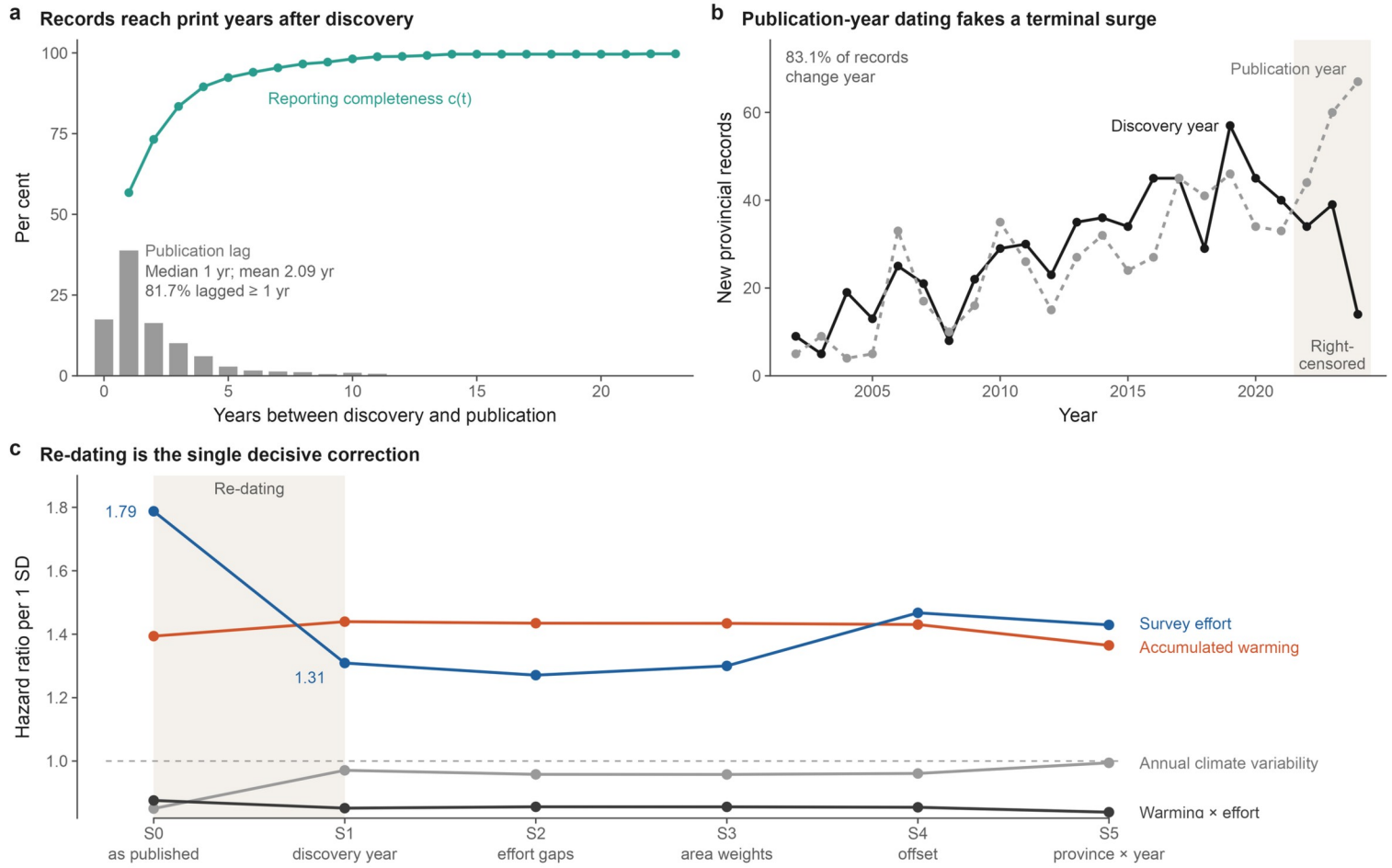


Fig. 2 Discovery vs publication dating

Backup

Specification robustness: climate indicator, window and effort proxy

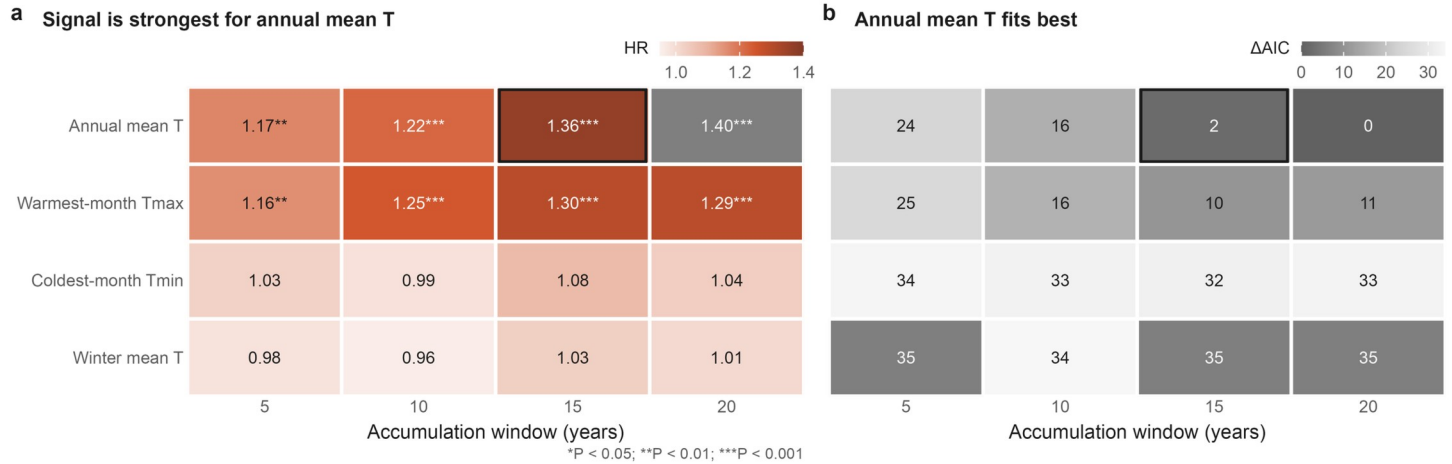


Fig. 3 Specification grid

Backup

Model anatomy: climate referenced to each species' range

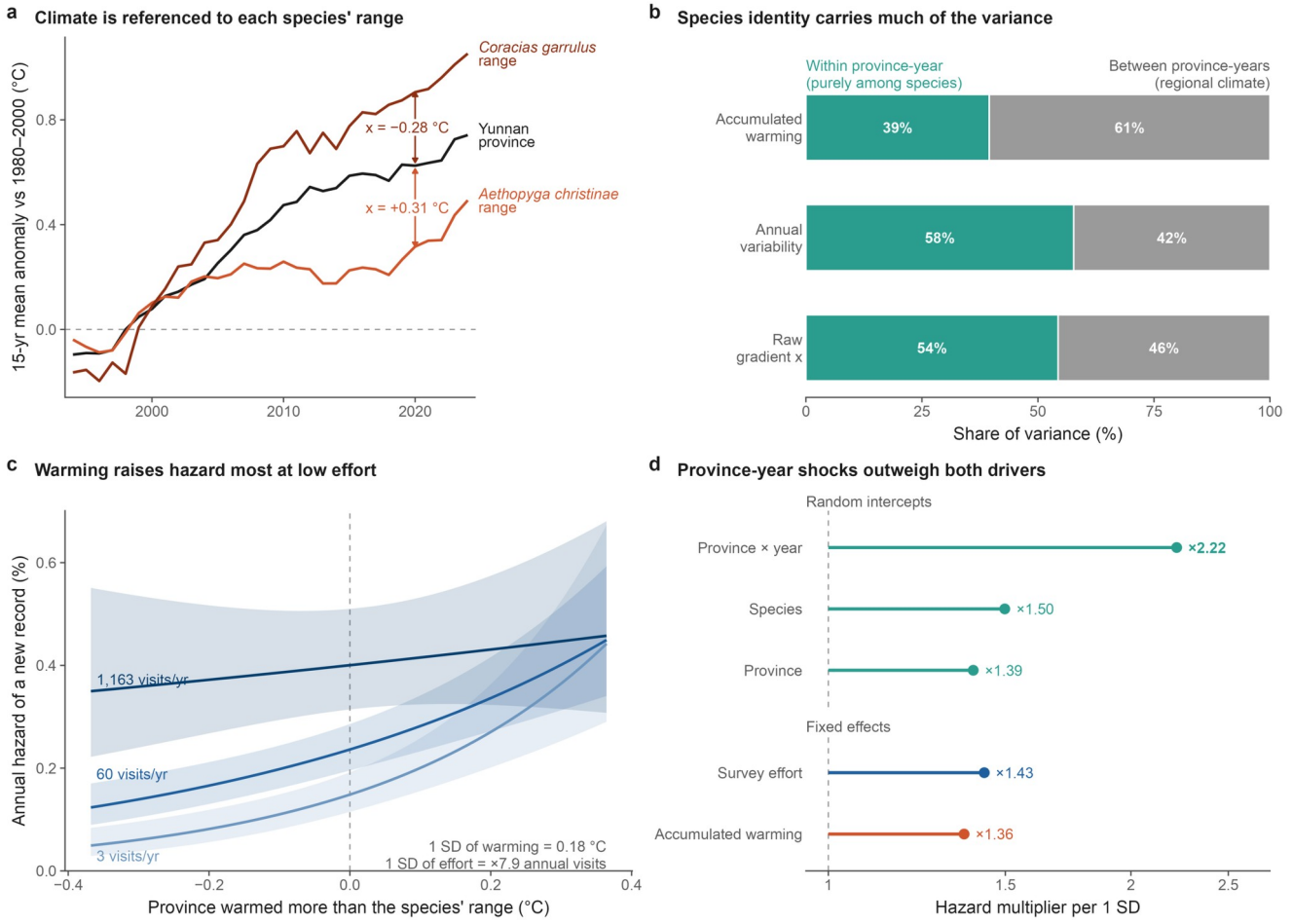
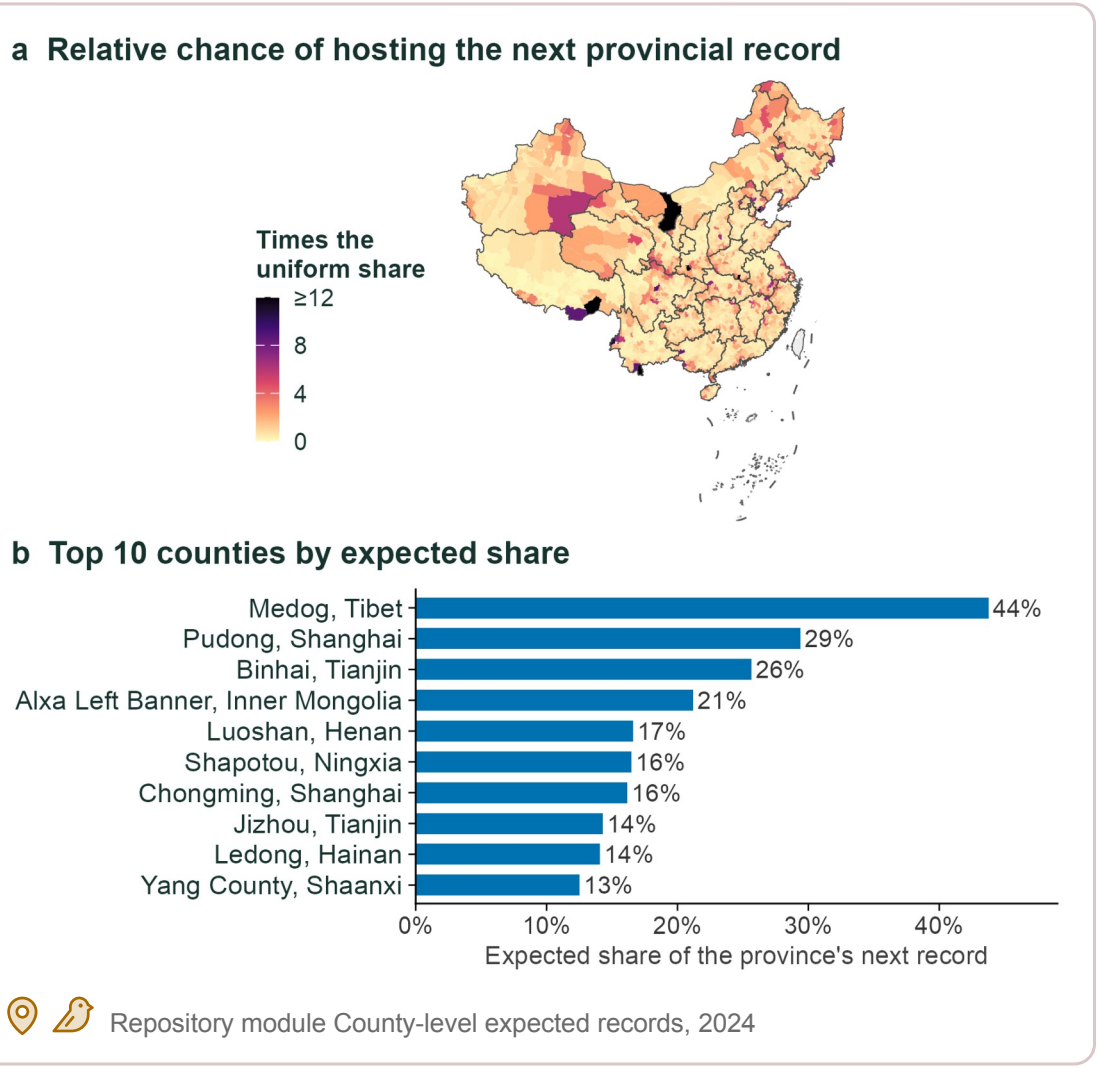


Fig. 4 Model anatomy

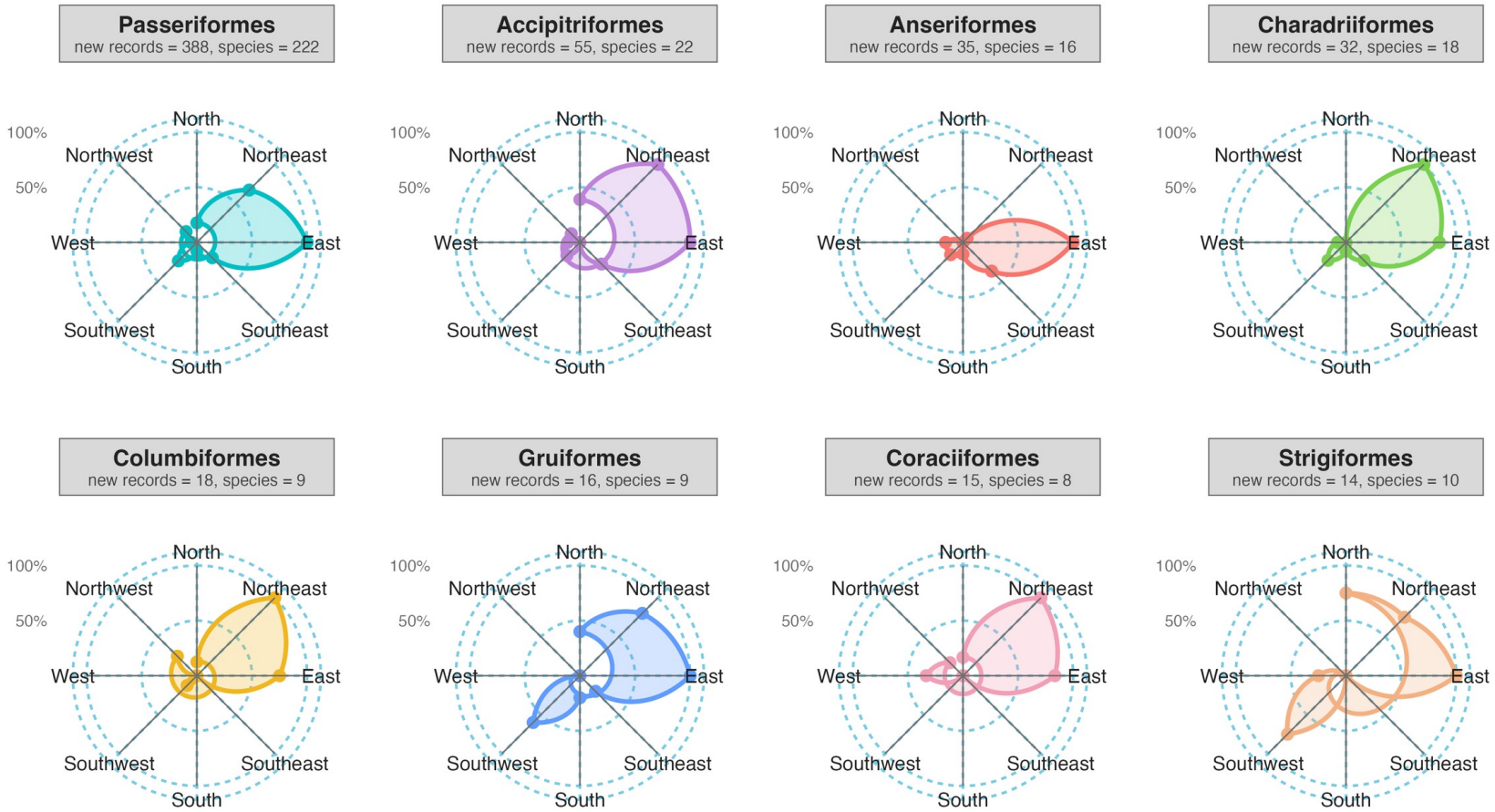
Backup

County-level surfaces: where to look first



Backup

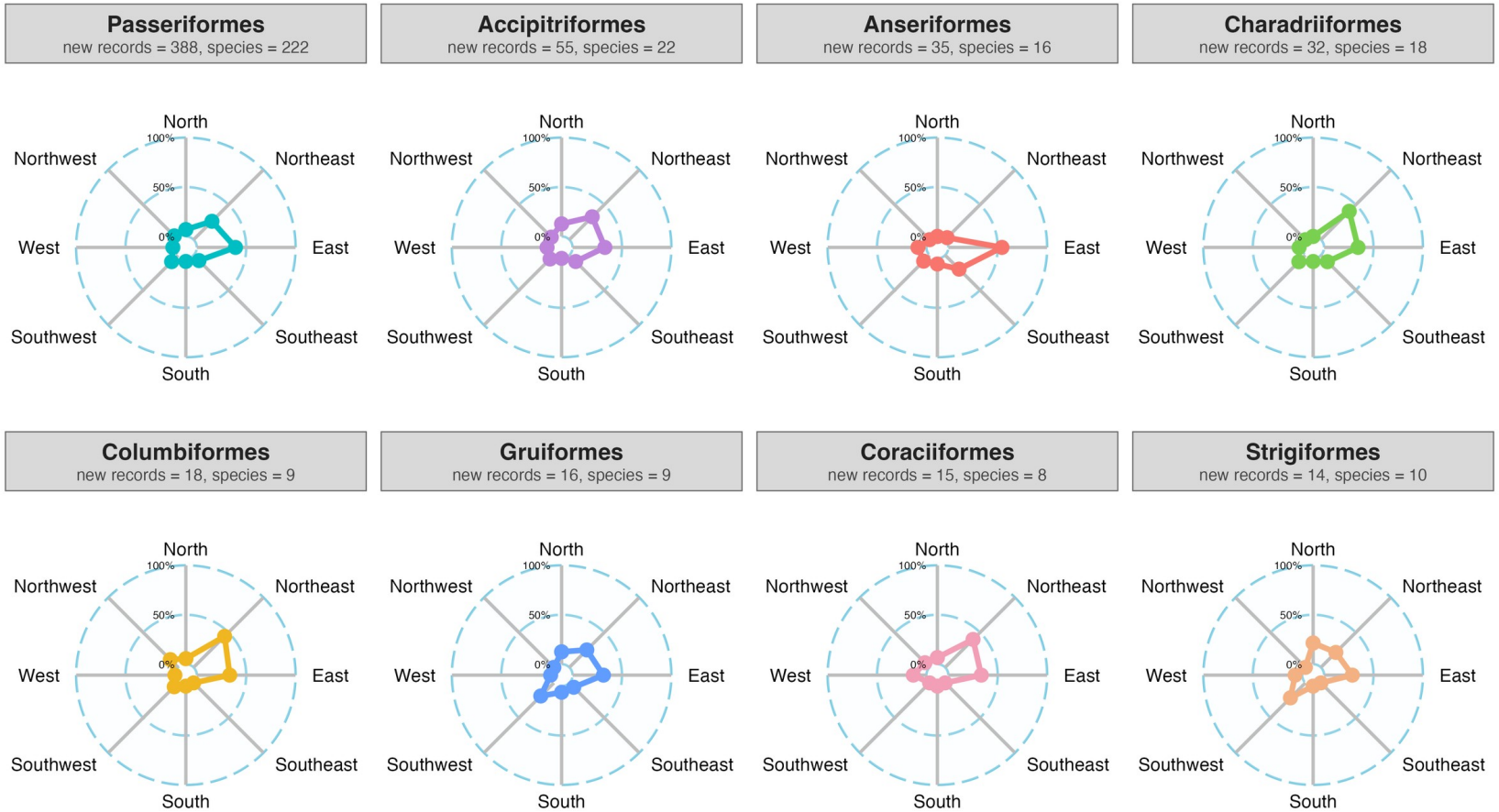
Direction by order: wind roses of the eight orders with most records



Share of each order's own records by compass sector

Backup

Direction by order: radar profiles of the eight orders with most records

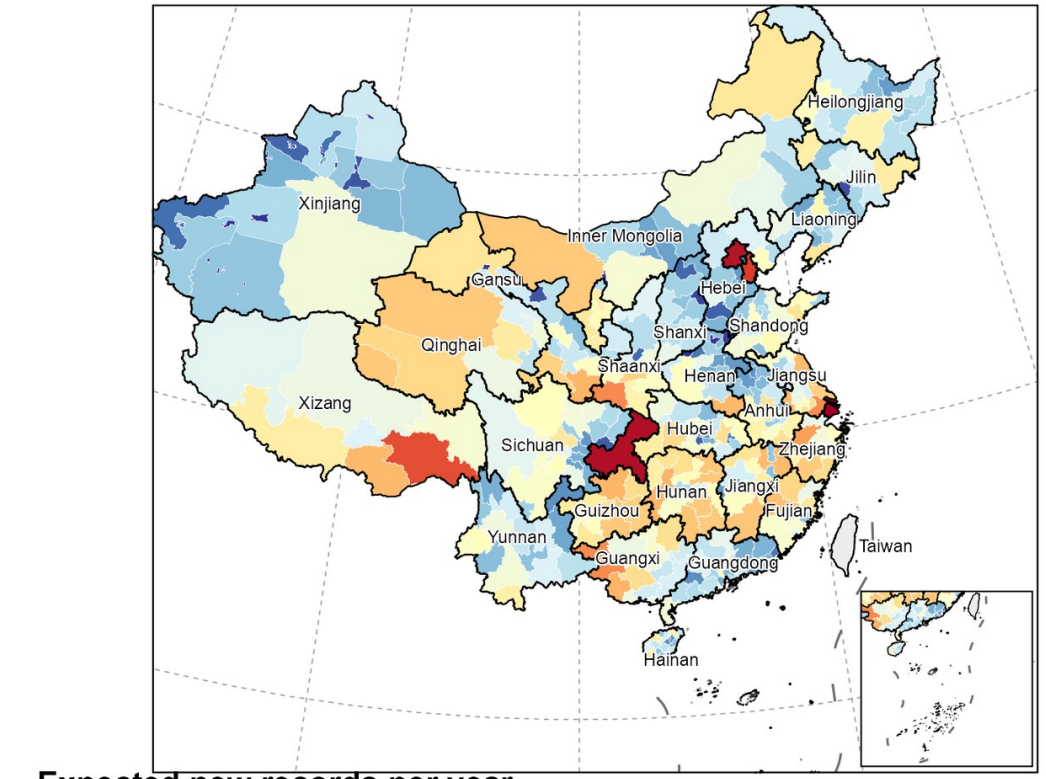


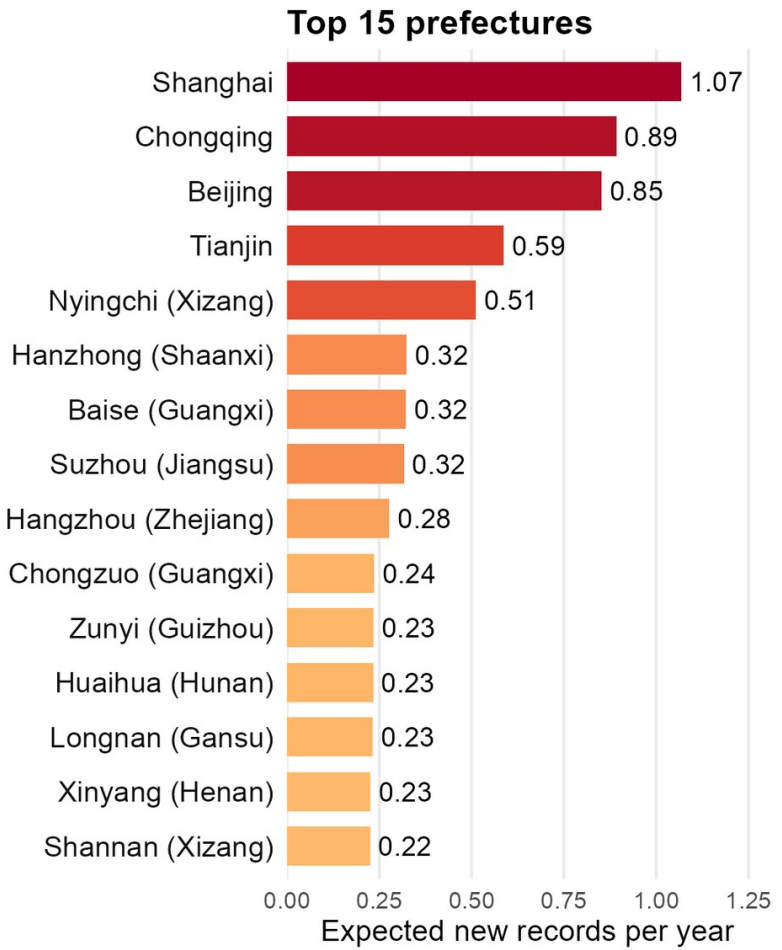
Radar profiles: share of each order's records by sector

Backup

Prefecture view of the 2030 futures: where to look first

2030 · SSP2-4.5 · effort growing: China 30.9 records per year





2030 · SSP2-4.5 · growing effort; prefecture shares held at 2024